

RESULTS OF PID CONTROLLER

First Lap:

```
[INFO] [1791451797.341800480] [lap_analyzer]: Lap Analyzer: Loaded path with 100 waypoints, perimeter: 528.20 m
[INFO] [1791451877.988427791] [lap_analyzer]: =====
[INFO] [1791451877.989533407] [lap_analyzer]: 🚩 LAP COMPLETED 🚩
[INFO] [1791451877.990446371] [lap_analyzer]: Lap Time: 80.47 s | Best Lap: 80.4 s
[INFO] [1791451877.991100908] [lap_analyzer]: Total Distance: 481.53 m
[INFO] [1791451877.991743998] [lap_analyzer]: -----
-----
[INFO] [1791451877.992395843] [lap_analyzer]: CTE -> Mean: 0.347 m | Max: 1.97 m | RMS: 0.491 m
[INFO] [1791451877.992945625] [lap_analyzer]: Speed -> Mean: 5.99 m/s | Max: 7.5 m/s
```

Second Lap :

```
[INFO] [1791451877.993424740] [lap_analyzer]: =====
[INFO] [1791451959.087678811] [lap_analyzer]: =====
[INFO] [1791451959.088292035] [lap_analyzer]: 🚩 LAP COMPLETED 🚩
[INFO] [1791451959.088838459] [lap_analyzer]: Lap Time: 81.13 s | Best Lap: 80.4 s
[INFO] [1791451959.089384647] [lap_analyzer]: Total Distance: 973.17 m
[INFO] [1791451959.089914573] [lap_analyzer]: -----
-----
[INFO] [1791451959.090451490] [lap_analyzer]: CTE -> Mean: 0.437 m | Max: 2.75 m | RMS: 0.639 m
[INFO] [1791451959.090972179] [lap_analyzer]: Speed -> Mean: 6.06 m/s | Max: 7.5 m/s
[INFO] [1791451959.091510316] [lap_analyzer]: =====
[INFO] [1791452037.787116400] [lap_analyzer]: =====
```

Third Lap :

```
[INFO] [1791451959.091510316] [lap_analyzer]: =====
[INFO] [1791452037.787116400] [lap_analyzer]: =====
[INFO] [1791452037.787758582] [lap_analyzer]: 🚩 LAP COMPLETED 🚩
[INFO] [1791452037.788305636] [lap_analyzer]: Lap Time: 78.68 s | Best Lap: 78.68 s
[INFO] [1791452037.788832031] [lap_analyzer]: Total Distance: 1458.45 m
[INFO] [1791452037.789352068] [lap_analyzer]: -----
-----
[INFO] [1791452037.789877281] [lap_analyzer]: CTE -> Mean: 0.397 m | Max: 1.941 m | RMS: 0.545 m
[INFO] [1791452037.790403672] [lap_analyzer]: Speed -> Mean: 6.17 m/s | Max: 7.30 m/s
[INFO] [1791452037.790904987] [lap_analyzer]: =====
[INFO] [1791452117.385979183] [lap_analyzer]: =====
```

RESULTS OF PURE PURSUIT

First Lap:

```
1 waypoints, perimeter: 528.20 m
[INFO] [1791452480.447773598] [lap_analyzer]: =====
[INFO] [1791452480.448027755] [lap_analyzer]: 🚩 LAP COMPLETED 🚩
[INFO] [1791452480.448237390] [lap_analyzer]: Lap Time: 71.41 s | Best Lap: 71.4
1 s
[INFO] [1791452480.448427911] [lap_analyzer]: Total Distance: 444.49 m
[INFO] [1791452480.448616666] [lap_analyzer]: -----
-----
[INFO] [1791452480.448808109] [lap_analyzer]: CTE -> Mean: 0.030 m | Max: 0.31
7 m | RMS: 0.052 m
[INFO] [1791452480.448996493] [lap_analyzer]: Speed -> Mean: 6.24 m/s | Max: 8.1
7 m/s
[INFO] [1791452480.449195300] [lap_analyzer]: =====
```

Second Lap :

```
[INFO] [1791452480.449195300] [lap_analyzer]: =====
[INFO] [1791452549.848173588] [lap_analyzer]: =====
[INFO] [1791452549.848539000] [lap_analyzer]: 🚩 LAP COMPLETED 🚩
[INFO] [1791452549.848858494] [lap_analyzer]: Lap Time: 69.40 s | Best Lap: 69.4
0 s
[INFO] [1791452549.849250734] [lap_analyzer]: Total Distance: 888.96 m
[INFO] [1791452549.849569330] [lap_analyzer]: -----
-----
[INFO] [1791452549.849890912] [lap_analyzer]: CTE -> Mean: 0.029 m | Max: 0.31
4 m | RMS: 0.051 m
[INFO] [1791452549.850185868] [lap_analyzer]: Speed -> Mean: 6.40 m/s | Max: 8.0
5 m/s
[INFO] [1791452549.850477625] [lap_analyzer]: =====
```

Third Lap :

```
[INFO] [1791452549.850477625] [lap_analyzer]: =====
[INFO] [1791452619.645418873] [lap_analyzer]: =====
[INFO] [1791452619.645675725] [lap_analyzer]: 🚩 LAP COMPLETED 🚩
[INFO] [1791452619.645869510] [lap_analyzer]: Lap Time: 69.80 s | Best Lap: 69.4
0 s
[INFO] [1791452619.646074519] [lap_analyzer]: Total Distance: 1333.62 m
[INFO] [1791452619.646274487] [lap_analyzer]: -----
-----
[INFO] [1791452619.646472689] [lap_analyzer]: CTE -> Mean: 0.032 m | Max: 0.30
9 m | RMS: 0.056 m
[INFO] [1791452619.646721877] [lap_analyzer]: Speed -> Mean: 6.37 m/s | Max: 7.7
8 m/s
[INFO] [1791452619.646918269] [lap_analyzer]: =====
```

RESULTS OF MPC

First Lap :

```
[INFO] [1791452693.279304602] [lap_analyzer]: Initializing Lap Analyzer Node...
[INFO] [1791452695.591326171] [lap_analyzer]: Lap Analyzer: Loaded path with 100
1 waypoints, perimeter: 528.20 m
[INFO] [1791452759.001061274] [lap_analyzer]: =====
[INFO] [1791452759.001312032] [lap_analyzer]: 🏁 LAP COMPLETED 🏁
[INFO] [1791452759.001507019] [lap_analyzer]: Lap Time: 63.51 s | Best Lap: 63.5
1 s
[INFO] [1791452759.001695182] [lap_analyzer]: Total Distance: 446.15 m
[INFO] [1791452759.001881119] [lap_analyzer]: -----
-----
[INFO] [1791452759.002072865] [lap_analyzer]: CTE -> Mean: 0.068 m | Max: 0.34
9 m | RMS: 0.094 m
[INFO] [1791452759.002278079] [lap_analyzer]: Speed -> Mean: 7.04 m/s | Max: 7.2
9 m/s
```

Second Lap :

```
[INFO] [1791452759.002278079] [lap_analyzer]: Speed -> Mean: 7.04 m/s | Max: 7.2
9 m/s
[INFO] [1791452759.002467562] [lap_analyzer]: =====
[INFO] [1791452821.489758486] [lap_analyzer]: =====
[INFO] [1791452821.489993068] [lap_analyzer]: 🏁 LAP COMPLETED 🏁
[INFO] [1791452821.490222756] [lap_analyzer]: Lap Time: 62.45 s | Best Lap: 62.4
5 s
[INFO] [1791452821.490435858] [lap_analyzer]: Total Distance: 892.82 m
[INFO] [1791452821.490632018] [lap_analyzer]: -----
-----
[INFO] [1791452821.490818601] [lap_analyzer]: CTE -> Mean: 0.075 m | Max: 0.36
1 m | RMS: 0.104 m
[INFO] [1791452821.491003067] [lap_analyzer]: Speed -> Mean: 7.15 m/s | Max: 7.2
2 m/s
```

Third Lap :

```
[INFO] [1791452821.491195271] [lap_analyzer]: =====
[INFO] [1791452883.889943932] [lap_analyzer]: =====
[INFO] [1791452883.890196896] [lap_analyzer]: 🏁 LAP COMPLETED 🏁
[INFO] [1791452883.890396991] [lap_analyzer]: Lap Time: 62.44 s | Best Lap: 62.4
4 s
[INFO] [1791452883.890600970] [lap_analyzer]: Total Distance: 1338.76 m
[INFO] [1791452883.890790109] [lap_analyzer]: -----
-----
[INFO] [1791452883.891013194] [lap_analyzer]: CTE -> Mean: 0.073 m | Max: 0.51
9 m | RMS: 0.107 m
[INFO] [1791452883.891239008] [lap_analyzer]: Speed -> Mean: 7.15 m/s | Max: 7.2
1 m/s
[INFO] [1791452883.891453130] [lap_analyzer]: =====
```

